

# **EC-201: Engineering Project Management**

## **Project Selection**

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## 8.1 Introduction (1/2)

- Project selection is the *process of choosing a project or set of projects to be implemented by the organization*. Since projects in general require a substantial investment in terms of money and resources, both of which are limited, it is of vital importance that the projects that an organization selects provide good returns on the resources and capital invested.
- Because a successful model must capture every critical aspect of the decision, more complex decisions typically require more sophisticated models.
- Project decisions can impact every business stakeholder, including customers, employees, partners, regulators, and shareholders. A sophisticated model may be needed to capture strategic suggestions.

## 8.1 Introduction (2/2)

- Project decisions typically produce many different types of impacts on the organization. Making good decisions requires not just estimating the financial return on investment; it requires understanding all of the ways that projects add value. A more sophisticated model is needed to account for all of the different types of potential impacts that project selection decisions can create.

## 8.2 Project Decisions (1 / 3)

- Project decisions often entail risk and uncertainty. More sophisticated model is needed to correctly deal with risk and uncertainty.
- Project selection is the process of evaluating individual projects or groups of projects, and then choosing to implement some set of them so that the objectives of the parent organization will be achieved. This same systematic process can be applied to any area of the organization's business in which choices must be made between competing alternatives. For example:
  - A manufacturing firm can use evaluation/selection techniques to choose which machine to adopt in a part-fabrication process.
  - A hospital can find the best mix of psychiatric, orthopaedic, obstetric, and other beds for a new wing.

## 8.2 Project Decisions (2/3)

- Each project will have different costs, benefits, and risks. Rarely are these known with certainty.
- We will discuss several techniques that can be used to help senior managers select projects. Project selection is only one of many decisions associated with project management.
- To deal with all of these problems, we use *decision aiding models*.
- Reality is far too complex to deal with in its wholeness. An “idealist” is needed to reveal almost all the reality from a problem, leaving only the aspects of the “real” situation with which he or she wishes to deal. The idealized version of the problem that results is called a model.
- The model represents the problem’s *structure*, its form.

## 8.2 Project Decisions (3/3)

- Models may be quite simple to understand, or they may be extremely complex. In general, introducing more reality into a model tends to make the model more difficult to manipulate. If the input data for a model are not known precisely, we often use probabilistic information; that is, the model is said to be *stochastic* rather than *deterministic*.
- **Modelling-** *“A model is an explicit statement of our image of reality. It is a representation of the relevant aspects of the decision with which we are concerned. It represents the decision area by structuring and formalizing the information we possess about the decision and, in doing so, presents reality in a simplified organized form. A model, therefore, provides us with an abstraction of a more complex reality”.*  
(Cooke and Slack, 1991)

## 8.3 Criteria for Choosing Project Model (1/3)

When a firm chooses a project selection model, the following criteria, based on Souder (1973), are most important:

- 1. *Realism***- The model should reflect the reality of the manager's decision situation, including the multiple objectives of both the firm and its managers. Without a common measurement system, direct comparison of different projects is impossible. For example, Project A may strengthen a firm's market share by extending its facilities, and Project B might improve its competitive position by strengthening its technical staff. The model should also include factors that reflect project risks, including the technical risks of performance, cost, and time as well as the market risks of customer rejection and other implementation risks.



## 8.3 Criteria for Choosing Project Model (2/3)

2. *Capability-* The model should be sophisticated enough to deal with multiple time periods, simulate various situations both internal and external to the project (for example, strikes, interest rate changes), and optimize the decision.
3. *Flexibility-* The model should give valid results within the range of conditions that the firm might experience. It should have the ability to be easily modified, or to be self-adjusting in response to changes in the firm's environment; for example, tax laws change, new technological advancements alter risk levels, and, above all, the organization's goals change.
4. *Ease of Use-* The model should be reasonably convenient, not take a long time to execute, and be easy to use and understand.

## 8.3 Criteria for Choosing Project Model (3/3)

5. *Cost-* Data gathering and modelling costs should be low relative to the cost of the project and must surely be less than the potential benefits of the project.
6. *Easy Computerization-* It should be easy and convenient to gather and store the information in a computer database, and to manipulate data in the model through use of widely available, standard computer packages such as Excel, Visio, and like programs. The same ease and convenience should apply to transferring the information to any standard decision support system.

## 8.4 The Nature of Project Selection Models (1/3)

- There are two basic types of project selection models, **numeric** and **nonnumeric**. Both are widely used. Many organizations use both at the same time, or they use models that are combinations of the two. Nonnumeric models, as the name implies, do not use numbers as inputs. Numeric models do, but the criteria being measured may be either objective or subjective.
- Before examining specific kinds of models within the two basic types, let us consider just what we wish the model to do for us, never forgetting two critically important, but often overlooked facts.
  - Models do not make decisions-people do. The manager, not the model, bears responsibility for the decision.
  - All models, however sophisticated, are only partial representations of the reality they are meant to reflect. Reality is far too complex for us to capture more than a small fraction of it in any model. Therefore, no model can yield an optimal decision except within its own, possibly inadequate, framework.

## 8.4 The Nature of Project Selection Models (2/3)

- We seek a model to assist us in making project selection decisions. It should evaluate potential projects by the degree to which they will meet the firm's objectives. To construct a selection/evaluation model, therefore, it is necessary to develop a list of the firm's objectives.
- A list of objectives should be generated by the organization's top management.
- When the list of objectives has been developed, an additional refinement is recommended. The elements in the list should be *weighted*. Each item is added to the list because it represents a contribution to the success of the organization, but each item does not make an equal contribution. The weights reflect different degrees of contribution each element makes in accomplishing a set of goals.

## 8.4 The Nature of Project Selection Models (3/3)

- Once the list of goals has been developed, one more task remains. The probable contribution of each project to each of the goals should be estimated. A project is selected or rejected because it is predicted to have certain outcomes if implemented.
- These outcomes are expected to contribute to goal achievement. If the estimated level of goal achievement is sufficiently large, the project is selected. If not, it is rejected.

## 8.5 Types of Project Selection Models (1 / 4)

Of the two basic types of selection models (numeric and nonnumeric), nonnumeric models are older and simpler and have only a few subtypes to consider. We examine them first.

- ***Non-Numeric Models:***

1. ***The Sacred Cow-*** In this case the project is suggested by a senior and powerful official in the organization. Often the project is initiated with a simple comment such as, “If you have a chance, why don’t you look into . . .,”. The immediate result of this bland statement is the creation of a “project” to investigate whatever the boss has suggested. The project is “sacred” in the sense that it will be maintained until successfully concluded, or until the boss, personally, recognizes the idea as a failure and terminates it.
2. ***Capability-*** If a flood is threatening the plant, a project to build a protective dike (dam) does not require much formal evaluation, which is an example of this scenario.

## 8.5 Types of Project Selection Models (2/4)

3. *The Competitive Necessity-* Using this criterion, XYZ Steel undertook a major plant rebuilding project in the late 1960s in its steel bar manufacturing facilities near Chicago. It had become apparent to XYZ's management that the company's bar mill needed modernization if the firm was to maintain its competitive position in the Chicago market area. In a similar manner, many business schools are restructuring their undergraduate and Masters in Business Administration (MBA) programs to stay competitive with the more forward looking schools. In large part, this action is driven by declining numbers of tuition paying students and the need to develop stronger programs to attract them.
4. *The Product Line Extension-* In this case, a project to develop and distribute new products would be judged on the degree to which it fits the firm's existing product line, fills a gap, strengthens a weak link, or extends the line in a new, desirable direction.

## 8.5 Types of Project Selection Models (3/4)

5. *Comparative Benefit Model* - For this situation, assume that an organization has many projects to consider, perhaps several dozen. Senior management would like to select a subset of the projects that would most benefit the firm, but the projects do not seem to be easily comparable. For example, some projects concern potential new products, some concern changes in production methods, others concern computerization of certain records, and still others cover a variety of subjects not easily categorized. The organization has no formal method of selecting projects, but members of the selection committee think that some projects will benefit the firm more than others, even if they have no precise way to define or measure “benefit.”



## 8.5 Types of Project Selection Models (4/4)

6. *Q-Sort Model*- Of the several techniques for ordering projects, the Q-Sort is one of the most straightforward. First, the projects are divided into three groups *good, fair, and poor* according to their relative merits. If any group has more than eight members, it is subdivided into two categories, such as *fair-plus and fair-minus*. When all categories have eight or fewer members, the projects within each category are ordered from best to worst. Again, the order is determined on the basis of relative merit. The rater may use specific criteria to rank each project, or may simply use general overall judgment. Projects can then be selected in the order of preference, though they are usually evaluated financially before final selection.

## References:

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